

Perform Z-test and t-test for single population mean.

Problem Statement

A company produces rice bags and claims that the average weight is 50 kg. The population standard deviation of the bag weights is known to be 1.2 kg. A random sample of 30 bags from Factory A has a mean weight of 49.2 kg.

The company also sources rice bags from a second factory. To check whether the two factories produce bags with significantly different average weights, an inspector collects the following data:

Factory	Sample Size	Sample Mean	Population Std. Dev.
Factory A	30	49.5 kg	1.2 kg
Factory B	35	50.5	1.4 kg

You are required to perform hypothesis testing using Python.

1. One-Sample Z-Test

Test whether Factory A's rice bags provide sufficient evidence to reject the company's claim that the average bag weight is 50 kg.

2. Two-Sample Z-Test

Test whether there is a statistically significant difference between the average weights of rice bags produced by Factory A and Factory B.

3. For each test, report:

- Null hypothesis (H_0)
- Alternative hypothesis (H_1)
- Significance level ($\alpha = 0.05$)
- Z-statistic
- p-value
- Decision: Reject or Fail to Reject H_0
- Final conclusion in the context of the problem

4. Python Implementation

Use Python to perform the calculations. Display the test statistic and p-value and clearly interpret the result.